From Slow Reasoning to Real-Time, Reusable Robot Capabilities

Yixuan Huang

1 Introduction

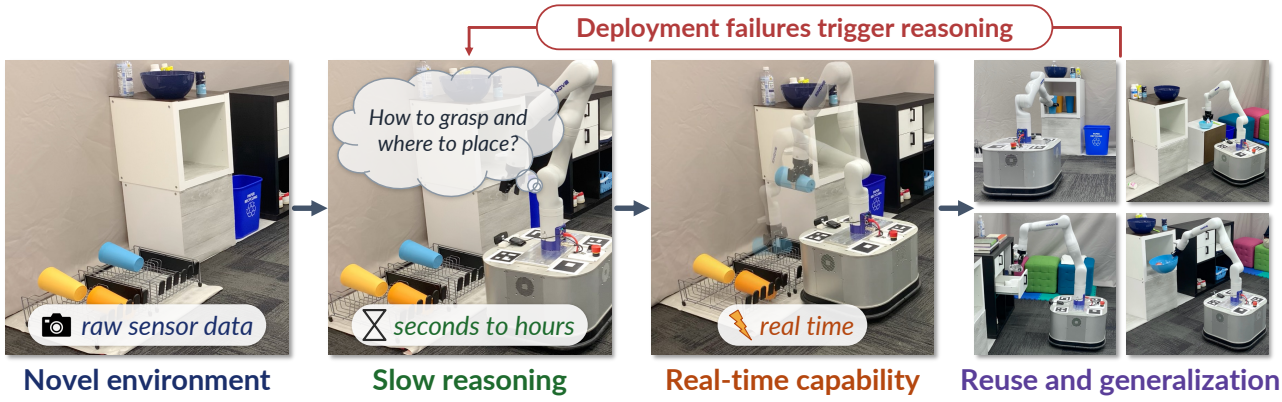


Figure 1: My research aims to enable robots to solve novel physical problems through slow reasoning, distill solutions into real-time, reusable visuomotor policies, and generalize these capabilities to new tasks and environments. Deployment failures trigger further reasoning, allowing robots to continually expand what they can do.

I aim to build robots that continually expand their **physical capabilities through experience**: solve unfamiliar tasks, acquire reusable capabilities, and adapt to new environments, embodiments, and users. Recent advances in robot learning have enabled increasingly capable visuomotor policies that can map visual observations directly to robot actions, enabling fast execution that responds to environmental changes. Despite this progress, collecting robot data that covers the diversity of physical interactions, objects, and environments remains prohibitively expensive, and pretrained policies still struggle to generalize to novel environments and tasks not covered by the training data [1, 2]. Building general-purpose robots therefore requires more than executing learned policies: **robots must also find ways to solve problems beyond their existing capabilities.**

Reasoning provides a complementary approach to learned policies. Task and motion planning (TAMP) [3, 4] exemplifies this process by interleaving symbolic planning and continuous geometric planning. Recent coding agents [5] offer another example by generating programs, invoking specialized tools, testing solutions, and iteratively refining strategies. However, the reasoning can take seconds to hours and is therefore impractical to invoke continuously during physical interaction. **My research aims to bridge this gap by turning slow reasoning into real-time, reusable capabilities that let robots respond to physical changes.**

My past work has explored the key components of this reasoning-to-capability cycle. First, I developed **the first methods for learning abstract world models that enable hierarchical planning directly from partial-view point clouds** [6–8], enabling reasoning on real robots directly from raw sensory observations. Second, I developed methods that make reasoning more efficient by distilling reasoning directly into fast vision-language-action models (VLAs) that can be reused for future tasks [11, 12], and by learning score functions to prioritize promising plans, improving planning efficiency and success rates [9, 10]. Third, I developed systems that identify failures as signals for improving future capabilities [13, 14]. Together, these projects establish the foundations of my research: enabling robots to reason about novel physical problems, turn the resulting solutions into real-time, reusable capabilities, and improve those capabilities through experience.

Looking forward, I aim to develop robots that continually expand their capabilities by turning slow reasoning into real-time, reusable robot capabilities. I organize this agenda around four complementary research directions: (1) structured robot memory for maintaining decision-relevant physical knowledge, (2) uncertainty-aware reasoning for making reliable decisions in unfamiliar environments, (3) generalizing capabilities across diverse embodiments by learning transferable physical knowledge, and (4) specializing generalized capabilities to individual users through a personalization layer.

2 Past Work

2.1 Enabling Reasoning from Raw Sensor Data

To reason about physical problems, robots should predict how their actions will change the environment and use these predictions to plan toward a goal, despite having only partial, noisy sensor observations. To address this challenge, I developed **the first methods for learning abstract, object-centric world models to enable hierarchical planning from partial-view point clouds** [6–8, 15].

A key question I addressed is how to learn a representation directly from partial-view point clouds that can capture both spatial relations (e.g., the book is in the drawer) and temporal effects of robot actions (e.g., which relations would change if the robot sweeps the trash on the table). I addressed this question by integrating spatial-temporal reasoning over a learned latent representation from sensory data shown in Fig. 2. This representation captures both geometric and symbolic effects of actions within a shared latent space, and the learned latent dynamics model can predict future latent states to enable planning. I integrated these learned world models with hierarchical planning to enable robots to solve long-horizon manipulation tasks directly from partial-view point clouds. Given a language instruction and a partial scene observation, my approach uses sampling-based planning to achieve the goal while accounting for geometric constraints and object interactions. I demonstrated this capability on a real mobile manipulator across diverse tasks, including placing multiple objects in a geometrically constrained cupboard (Fig. 2). In real-world evaluations, my approach **successfully solved 86.7%** of the long-horizon tasks [8], demonstrating its capability in complex environments.

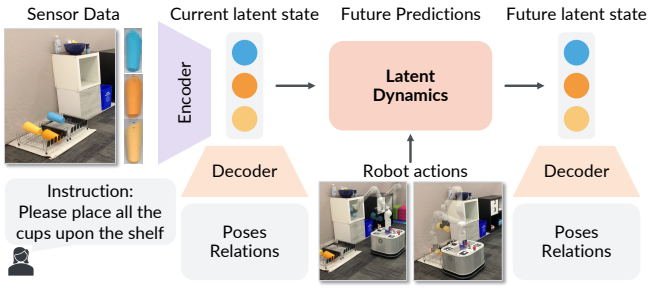


Figure 2: From partial-view point clouds, the robot learns an object-centric world model that predicts how robot actions change the environment, enabling goal-directed planning directly from raw sensory observations.

2.2 Turning Reasoning into Real-Time, Reusable Capabilities

Reasoning is powerful but expensive. Rather than repeatedly paying this cost, I have explored two complementary ways to make reasoning real-time and reusable: distilling reasoning to generate data into reusable visuomotor policies, and learning from previous experience to make future planning faster.

First, with my mentees Samrat and Liang, we explored distilling slow reasoning into real-time, reusable capabilities. In DATAFARM [11], we used TAMP to generate demonstrations offline and train fast VLAs that execute reactively on the real robot. We found that planner-generated trajectories are ineffective unless their behavioral distribution is aligned with VLA pretraining data. Fine-tuning on these aligned demonstrations raises success from **8.3% to 56.7%, approaching the oracle baseline that uses human teleoperation (61.7%)**. TANDEM [12] extends this approach to behaviors beyond the planner's capabilities by selectively using human teleoperation. Together, these projects demonstrate how slow reasoning can be transformed into reactive policies that can be reused during real-time physical interaction.

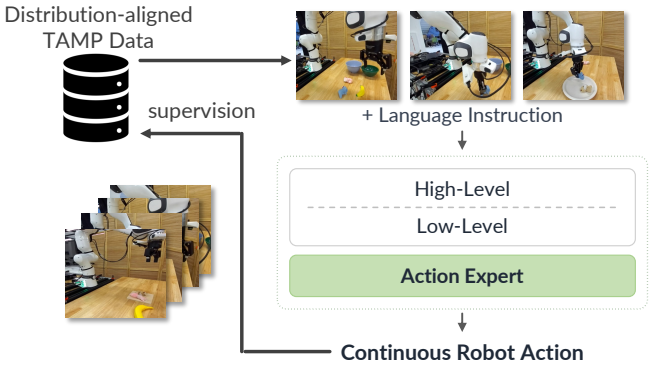


Figure 3: Distribution-aligned task and motion planning (TAMP) demonstrations provide supervision for fine-tuning downstream VLAs, transforming solutions generated through reasoning into reactive policies for real-time robot execution.

Second, my mentees Joseph, Qihang, and I developed methods that learn to accelerate reasoning itself. In SPECTRE [9], we developed a learned plan ranker that uses evidence from failed motion refinements to prioritize promising plans. In SC-BPE [10], we similarly learned to score candidate base placements jointly with plan skeletons before expensive refinement. Across eight task settings and two mobile manipulators, SC-BPE raises the average success rate from **54% to 89%** and reduces the

average planning time by **50%** compared with random ordering of the same candidates. **In these two projects, the robot reuses learned guidance to make subsequent planning more efficient.**

2.3 Learning from Failures to Expand Capabilities

Neither slow reasoning nor fast execution can guarantee reliable robot behavior in the complex real world. Even with abstract world models and real-time, reusable capabilities, robots can still fail in situations outside their training distribution. Rather than treating these failures as endpoints, I explored how robots can use them to identify capability gaps and improve their models. In Fail2Progress [13], I developed a failure-driven adaptation framework that detects failures during real-world execution, triggers recovery through replanning, generates targeted data around the observed failure, and uses this data to refine the robot's world model. This iterative loop of failure detection, recovery, and targeted model updates improves performance by **over 60%** in out-of-distribution environments, demonstrating the promise of refining robot models during deployment. Failure reasoning also benefits hierarchical systems built on foundation models. In collaborative work on routing deformable linear objects such as cables [14], a vision-language model plans over reinforcement-learned skills, detects execution failures such as unsuccessful clip insertions, and triggers a recovery skill before replanning, achieving **92%** success across long-horizon routing scenarios.



Figure 4: When unexpected failures reveal gaps in a robot's existing capabilities, the robot can diagnose the failure, collect targeted data, and update its models and policies. This feedback loop enables robots to improve their capabilities through real-world experience.

My work reframes failure as a trigger for capability expansion: when a real-time capability is insufficient, the robot can use reasoning to understand what went wrong, generate targeted experience, and update the capability so that similar failures become less likely in the future. **This provides the feedback loop in my broader research program: deployment failures can trigger reasoning that creates future reusable capabilities.**

3 Future Directions and Long-term Vision

My past work has explored how robots can build structured representations for reasoning, make reasoning more efficient, distill its solutions into real-time capabilities, and learn from failures during deployment. Building on these foundations, I will prioritize the following future directions.

Structured Robot Memory. Robots operating in partially observable environments must reason about objects that are temporarily occluded or outside their current field of view. Doing so requires them to retain and use information from past observations. In my prior work [15], I combined learned world models with video tracking to reason about unobserved objects, leveraging the models' ability to predict future states and maintain useful information beyond the current observation. However, maintaining memory alone is not enough: robots must decide which past observations remain useful for future decisions, especially as they encounter unfamiliar environments. I will investigate how robots can **selectively retain, update, or discard observations and object information based on their relevance to future decisions**, rather than storing all past experience indiscriminately. By preserving decision-relevant knowledge across interactions, robots can build on accumulated experience rather than repeatedly reconstructing their understanding of the environment, making future reasoning more efficient and reliable. The goal is to enable robots to maintain decision-relevant physical knowledge that supports reliable reasoning and planning under partial observability.

Uncertainty-aware Reasoning. Physical interactions are inherently uncertain: contact, friction, and object dynamics can vary across environments, making the outcomes of robot actions difficult to predict reliably. My prior work [16] studied probabilistic models of manipulation effects, moving beyond single-point predictions to represent distributions over possible outcomes. Building on this foundation, I will investigate how to model and quantify uncertainty in physical interactions and translate uncertainty estimates into reliable decisions. A

key challenge is that uncertainty estimates derived from learned models may not remain reliable in unfamiliar physical environments. I will therefore study **how robots can use execution outcomes, including discrepancies between predicted and observed effects, to calibrate their uncertainty estimates during deployment and improve decisions about when to act, gather additional observations, or invoke more expensive reasoning**. Ultimately, I aim to enable robots to make reliable decisions under uncertainty by identifying when existing capabilities are sufficient and when additional observations or more expensive reasoning are needed.

Generalizing Capabilities Across Embodiments. A key challenge in building general-purpose robots is enabling capabilities acquired by one robot to transfer to others with different physical embodiments. My prior work has demonstrated that slow reasoning can provide solutions across multiple hardware platforms (two fixed-base manipulators [6, 11], four mobile manipulators [1, 8, 10, 13], and a surgical robot [17]), given appropriate perception and kinematic models, but its computational cost makes it impractical for real-time interaction. Building on DATAFARM [11], I envision **using reasoning to generate training data across diverse robot platforms and fine-tune downstream VLAs**. However, generating data across embodiments alone does not guarantee that models will learn transferable physical knowledge rather than memorize embodiment-specific behaviors. My prior work has explored visual reasoning across diverse embodiments through RoboVista [18] and demonstrated zero-shot cross-embodiment transfer in VLA through LAP [19]. Building on these complementary foundations, I aim to investigate how slow reasoning across embodiments can provide supervision for learning transferable physical knowledge, beyond the transfer capabilities demonstrated by existing models. In particular, I will study how models can learn which aspects of a solution generalize across embodiments and which require adaptation to embodiment-specific constraints, such as kinematics and action spaces. Ultimately, I seek to turn reasoning across embodiments into real-time capabilities that transfer across robots, reducing the need to solve similar physical problems from scratch on each new robot.

Specializing Capabilities to Users. Even general-purpose robots must adapt to individual preferences and habits to provide meaningful value in people's everyday lives. I envision an off-the-shelf **personalization layer built on top of generalized robot policies**, allowing robots to adapt their behavior to individual users without retraining their entire policies or compromising their generalization capabilities. Building on **Homeostasis at Home**, a project I am currently pursuing with my mentee Jingru, I will investigate how robots can represent persistent user preferences, infer when those preferences apply, and adapt their behavior as new feedback becomes available. A central challenge is to learn user-specific preference representations from everyday observations and sparse feedback, and use them to adapt generalized policies without compromising their transferable capabilities. Ultimately, I aim to enable robots to reuse generalized capabilities while tailoring their execution to individual users, allowing them to integrate naturally into diverse, personalized environments.

Long-Term Vision. My long-term vision is to build robots that continually expand what they can do, rather than remain limited by the capabilities acquired during training. I envision robots that can reason to solve novel problems, turn these solutions into real-time, reusable capabilities, and transfer them across embodiments, environments, and users. As robots encounter new challenges, their reasoning can serve as learning signals, helping them acquire capabilities that let them solve future problems faster. This creates a virtuous cycle in which reasoning expands what robots can do, while accumulated experience and reusable capabilities reduce their reliance on expensive reasoning for similar situations. Ultimately, I aim to make robots that can grow beyond their initial capabilities and operate reliably in the diverse, changing, and personalized environments of everyday life.

4 Funding and Collaboration

During my Ph.D., I had the opportunity to engage with DARPA-funded research activities and present my work directly to the DARPA TIAMAT program manager, which provided valuable exposure to federally funded robotics research. During my postdoc, I helped my advisor write a successful grant to Amazon Robotics and a proposal to Sony Research Awards that is currently under review. I am also helping prepare an NSF proposal, gaining experience in collaborative proposal writing. As a faculty member, I plan to seek research funding from federal agencies such as NSF, DARPA, and ONR. My research on reasoning and real-time, reusable robot capabilities aligns closely with programs such as the NSF Foundational Research in Robotics program, the DARPA TIAMAT program, and the ONR Young Investigator Program. I will also actively seek industry funding from NVIDIA, Google, Amazon Robotics, Sony, and Toyota Research Institute.

References

[1] **Yixuan Huang***, Bowen Li*, Vaibhav Saxena*, Yichao Liang, Utkarsh Aashu Mishra, Liang Ji, Lihan Zha, Jimmy Wu, Nishanth Kumar, Sebastian Scherer, Danfei Xu, Tom Silver. *KinDER: A Physical Reasoning Benchmark for Robot Learning and Planning*. Robotics: Science and Systems (RSS), 2026.

[2] Hao Wang, **Yixuan Huang**, Somil Bansal, Tom Silver. *Chaining Visuomotor Skills for Constrained Manipulation Through Perceptual Manipulation*. Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027.

[3] Caelan Reed Garrett, Rohan Chitnis, Rachel Holladay, Beomjoon Kim, Tom Silver, Leslie Pack Kaelbling, Tomás Lozano-Pérez. *Integrated Task and Motion Planning*. Annual Review of Control, Robotics, and Autonomous Systems (ARCRAS), 2021.

[4] **Yixuan Huang***, Benned Hedegaard*, Naman Shah, Siddharth Srivastava, George Konidaris, Tom Silver. *Learning By and For Task and Motion Planning: A Survey*. Submitted to The International Journal of Robotics Research (IJRR), 2026.

[5] Matteo Merler, Bowen Li, Josh Roy, Yichao Liang, Qianwei Wang, **Yixuan Huang**, Tom Silver. *Coding Agents for Generalized Task and Motion Planning Problems*. Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027.

[6] **Yixuan Huang**, Adam Conkey, Tucker Hermans. *Planning for Multi-Object Manipulation with Graph Neural Network Relational Classifiers*. IEEE International Conference on Robotics and Automation (ICRA), 2023.

[7] **Yixuan Huang**, Nichols Crawford Taylor, Adam Conkey, Weiyu Liu, and Tucker Hermans. *Latent Space Planning for Multi-Object Manipulation with Environment-Aware Relational Classifiers*. IEEE Transactions on Robotics (T-RO), 2024.

[8] **Yixuan Huang**, Christopher Agia, Jimmy Wu, Tucker Hermans, Jeannette Bohg. *Points2Plans: From Point Clouds to Long-Horizon Plans with Composable Relational Dynamics*. IEEE International Conference on Robotics and Automation (ICRA), 2025.

[9] Jiacheng Xu, **Yixuan Huang**, Tom Silver. *SPECTRE: Skeleton Priority Estimation Conditioned on TAMP Refinement Evidence*. Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027.

[10] Qihang Li, **Yixuan Huang**, Tom Silver, Peng Gao. *Skeleton-Conditioned Base Placement Evaluation for Long-Horizon Mobile Manipulation*. Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027.

[11] Samrat Sahoo, **Yixuan Huang**, Tom Silver. *DATAFARM: Distribution-Aligned Task and Motion Planning for Fine-Tuning Vision-Language-Action Models*. Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027.

[12] Samrat Sahoo*, Liang Ji*, Tom Silver, **Yixuan Huang**. *TANDEM: Task and Motion Planning with As-Needed Demonstrations for Efficient Vision-Language-Action Model Fine-tuning*. Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027.

[13] **Yixuan Huang**, Novella Alvina, Mohanraj Devendran Shanthi, Tucker Hermans. *Fail2Progress: Learning from Real-World Robot Failures with Stein Variational Inference*. Conference on Robot Learning (CoRL), 2025.

[14] Mingen Li, Houjian Yu, **Yixuan Huang**, Youngjin Hong, Hantao Ye, Changhyun Choi. *Hierarchical DLO Routing with Reinforcement Learning and In-Context Vision-Language Models*. IEEE International Conference on Robotics and Automation (ICRA), 2026. **Best Paper Award on Robot Learning–Finalist**

[15] **Yixuan Huang**, Jialin Yuan, Chanho Kim, Pupul Pradhan, Bryan Chen, Li Fuxin, Tucker Hermans. *Out of Sight, Still in Mind: Reasoning and Planning about Unobserved Objects with Video Tracking Enabled Memory Models*. IEEE International Conference on Robotics and Automation (ICRA), 2024.

[16] Siyeon Kim, Mohanraj Devendran Shanthi, **Yixuan Huang**, Tucker Hermans. *Learning Multimodal Probabilistic Models of Manipulation Skill Effects*. Workshop on Learning to Simulate Robot Worlds, Conference on Robot Learning (CoRL), 2025.

[17] **Yixuan Huang**, Michael Bentley, Tucker Hermans, Alan Kuntz. *Toward Learning Context-Dependent Tasks from Demonstration for Tendon-Driven Surgical Robots*. International Symposium on Medical Robotics (ISMR), 2021. **Best Paper Award–Finalist & Best Student Paper Award–Finalist**

[18] Shuangyu Xie, Kaiyuan Chen, Ziyang Chen, Simeon Adebola, **Yixuan Huang**, Zehan Ma, Tianshuang Qiu, Wentao Yuan, Dhruv Shah, Pannag R Sanketi, Ken Goldberg. *RoboVista: Evaluating Vision Language Models for Diverse Robot Applications*. Robotics: Science and Systems (RSS), 2026.

[19] Lihan Zha, Asher James Hancock, Mingtong Zhang, Tenny Yin, **Yixuan Huang**, Dhruv Shah, Allen Z. Ren, Anirudha Majumdar. *LAP: Language-Action Pre-training Enables Zero-Shot Cross-Embodiment Transfer*. Robotics: Science and Systems (RSS), 2026.